

PrismIQ Analytics Report

Dataset Intelligence & Data Quality Summary

Generated By:	PrismIQ
Generated At:	2026-07-16 17:32 UTC
Dataset Name:	customeriq demo 100k
Dataset Version:	Version 1
Quality Score:	100.00/100
Report ID:	ac2b7766-1516-48bd-8ccb-eea1e62cd2e2
User ID:	bbfca9e2-788f-4419-8ace-5999bd3cb798
Engine Version:	1.0
Report Schema:	1.0

Executive Summary

The dataset 'customeriq demo 100k' currently holds a quality health score of **100.00/100**. The active version contains a total of 0 missing data cells and 0 duplicate rows. This summary highlights structural anomalies, cleaning lineages, and AI-recommended preprocessing steps.

Dataset Information

Dataset Name:	customeriq demo 100k
Original Filename:	customeriq_demo_100k.xlsx
Description:	No description provided
File Size (Bytes):	8466922
Import Timestamp:	2026-07-16 17:26:00 UTC

Quality Metrics & Schema Profile

Column Name	Data Type	Missing	Null %
CustomerID	numeric	0	0.00%
CustomerName	categorical	0	0.00%
Gender	categorical	0	0.00%
Age	numeric	0	0.00%
Email	text	0	0.00%
Country	categorical	0	0.00%
City	categorical	0	0.00%
InvoiceNo	text	0	0.00%
PurchaseDate	datetime	0	0.00%
ProductCategory	categorical	0	0.00%
ProductName	categorical	0	0.00%
Quantity	numeric	0	0.00%
UnitPrice	numeric	0	0.00%
TotalAmount	numeric	0	0.00%

Cleaning Summary & History

Removed Rows	Duplicates	Missing Handled	Outliers Handled	Time
No cleaning jobs applied yet.				

Exploratory Data Analysis (EDA)

Column	Mean	Median	Std Dev
CustomerID	12499.36	12502.00	1443.97
Age	44.08	44.00	15.23
Quantity	3.00	3.00	1.41
UnitPrice	75.12	62.42	51.74
TotalAmount	225.45	160.88	201.47

AI Insights Observations

[OUT_003] Category: Risk	Severity: HIGH
Observation: Extraneous value outliers detected in numeric variables.	Confidence: 0.92%
Root Cause: Outliers were discovered beyond standard 1.5x Interquartile Ranges (IQR), specifically in: `TotalAmount` (5780 outliers, 5.8%), `UnitPrice` (5671 outliers, 5.7%).	
[CORR_004] Category: Correlation	Severity: MEDIUM
Observation: Strong linear relationship detected between numeric columns.	Confidence: 0.95%
Root Cause: Pearson correlation analysis indicates strong associations: `Quantity` & `TotalAmount` (r=0.52), `UnitPrice` & `TotalAmount` (r=0.77). Highly correlated variables move together systematically.	
[CARD_005] Category: Statistics	Severity: LOW
Observation: High cardinality in categorical data columns.	Confidence: 0.9%
Root Cause: Categorical parameters contain high ratios of unique values: `InvoiceNo` (100000 unique, 100.0%), implying they represent textual logs, indexes, or identifiers.	
[TEMP_006] Category: Timeline	Severity: INFO
Observation: Temporal timelines coverage loaded successfully.	Confidence: 0.95%
Root Cause: Datetime intervals spans detected: `PurchaseDate` spanning 900.0 days (from 2023-01-01T00:00:00 to 2025-06-19T00:00:00).	

Actionable Recommendations

Observation Link	Recommended Action	Actionability
Risk	Evaluate outliers. Outliers can represent measurement faults or rare valid events. Consider applying IQR trimming to isolate distribution metrics.	85%
Correlation	These variables exhibit high linear dependency. Consider removing one column or performing dimensionality reduction to avoid multicollinearity in downstream models.	70%
Statistics	Ensure these columns are standard strings. Consider group standardizations or mapping key features.	60%
Timeline	If running cyclic or seasonal forecasts, ensure this timeline span covers at least one complete calendar year or cyclic sales window.	50%

Appendix & Methodology Documentation

Quality Score Index is calculated as a weighted average of missing cell percentages and duplicate row ratios. Missing weights and duplicate weights default to 50.0% each, adjusted via server configuration parameters.

Engine Version: 1.0.0 | **Report Schema:** 1.0 | **Checksum Algorithm:** SHA-256